

ProofRoad

A truck drives only as fast as the road ahead has proved itself clear.

Smart India Hackathon 2026 | Problem statement 26007 | Safe and efficient operation of mine vehicles in fog | Ministry of Steel / NMDC Limited | Version 2

The idea in one breath

In fog, a truck should only go as fast as the distance it can prove is clear allows. Every piece of information (own radar, another truck's message, a fog sensor, a road gate) counts as proof only while it is fresh. Old or missing information never counts as "clear".

Analogy. Indian Railways lets one train into a track section at a time and treats "no signal" as "stop". ProofRoad does the same for mine haul roads, without a control room in the safety loop.

The problem

At Bailadila in monsoon, visibility on hilltop haul roads can fall to 3 to 5 m, while a loaded dumper can need many times that to stop. Today's two answers are both bad: run blind (safe only by luck) or stop everything (ore stops moving and shifts are lost). ProofRoad measures how much road is **proven** clear, sets the speed to match, and slows only the stretch that is actually risky.

The rule that keeps it safe

Fresh evidence can shorten the proven distance at once. Only fresh proof can lengthen it, and **unknown is never counted as clear**. If the network, camera, fog post or dashboard dies, the truck's own radar warning still works. Other parts can make a truck more careful, never less careful than its own radar says.

The pitch in one line

"We did not add another dashboard. We built a system that can explain exactly why the next metre is safe, and becomes conservative when it cannot."

What is new compared with the source designs

- Message freshness by age counters, so no shared clock is needed.
- Hazards leave the list only when a truck actually looks, never by a timer.
- Right-of-way comes from how much room each truck has left to stop, not from a fixed loaded-wins rule.
- Fog posts are audited by the cameras of trucks that drive past.
- A virtual-fleet simulation gives efficiency numbers that two model trucks cannot.
- **[NEW]** Phantom Obstacle view: a hazard the driver cannot see is drawn on the road strip with its distance.
- **[NEW]** A dedicated safety chip is a stated design rule: the radar warning never depends on the AI computer.
- **[NEW]** Four-sensor perception (radar, RGB, thermal, NIR) under one rule: any sensor can shorten the proven distance, only radar or fresh peer proof can lengthen it.
- **[NEW]** Security by message class: a message that adds caution is lightly trusted; one that relaxes a restriction (gate GO, fog-post clear, hazard cleared) needs a per-device signature and a local check.
- **[NEW]** Driver interface built for a driver who must watch the windshield: a synthetic-vision road strip, three alert channels, and alert discipline that keeps warnings rare.

What changed in v2

Version 2 folds in the few ideas from the AegisMesh and FogNet Sentinel documents that fit the fixed must-build list. Everything else in those documents was reviewed and deliberately left out (see the source page). Changed items are shaded.

Change	From	Where it lands	Extra build effort
Security and trust page: message classes, threat table, prototype-versus-production keys, corrected forged-message claim, forged-GO demo.	Team discussion	New security page, demo 5:15, flaws, Q&A	Small (logic and paper; no hardware)
Driver interface section: synthetic-vision road strip, windshield-first attention rule with three alert channels, alert discipline against warning fatigue.	Team discussion	New driver-interface page, demo 2:15 and 3:00, flaws, Q&A	Medium (UI and alert logic; vibration motor and LED bar)
Light as a signal, not a sensor: amber driver screen; gate lamps flashing amber = hold, steady green = go.	Team discussion	Driver interface page, gates	Near zero
Sensor fusion: radar + RGB + thermal + NIR, with a one-way rule (extra sensors can only shorten proven distance).	Your requirement; fusion rule is ours	New fusion page, demo 1:30, cost table	Medium (thermal module + NoIR camera + association code)
Dedicated safety MCU owns the radar loop, corridor math, watchdog and buzzer. Pi/laptop does camera, AI and dashboard only.	AegisMesh (Tier 1)	Truck brain, architecture page	Near zero (already planned)
Phantom Obstacle: a follower sees the hazard as a marker with distance on its road strip.	AegisMesh (SVPP)	Hazard list, driver screen, demo 3:00	Small (UI on existing data)
Fixed driver screen: state safe speed clear distance needed distance.	FogNet Sentinel (HMI)	Driver screen	Small
Near-miss log added to the replay black box.	Sentinel / FogNet 4.1	Advise and log, dashboard	Small
Emergency beacon carried by a supervisor, signed, bypasses the control room. Still optional.	AegisMesh	Build list, judge Q&A	Optional
Idle trucks as temporary fog observers: simulation-only, not a demo claim.	AegisMesh	Flaws table	None (not built)
Worst-case table for 3 to 5 m fog, and why radar-first beats thermal/LiDAR here.	Review of the problem statement	Decision page, flaws	None (paper)

Scope guard. Two small sensor modules were added (thermal and NIR), plus a vibration motor and a short LED bar; everything else is already in the plan, a screen change, or a paper change. NIR is the first thing dropped if the build falls behind. The must-build list and the cut list on the demo page are unchanged.

Four parts, one rule

Layer	What is in it
Ops dashboard (watches and replays only)	Closures, analytics, event replay, near-miss trends
Fog posts	Visibility, trend, own health
Blind-curve gates	One direction at a time
Shared mesh	Truck beacons, hazard list, gate tokens, signed emergency stop
Truck safety loop (works with no network)	Sense: radar first, IMU, camera SafeCorridor: proven vs needed distance Advise and log: speed, reason, replay

The truck safety loop owns the safety decision. Everything else adds evidence and can only add caution.

Part	What it does
Truck brain (two chips)	[NEW] Safety MCU (STM32/ESP32): reads radar, runs the SafeCorridor formula, runs the alert logic and drives the buzzer, vibration motor and LED bar, and has a hardware watchdog. It has no dependency on anything else. Assist computer (Pi or laptop): RGB, thermal and NIR cameras, object naming, fog check, dashboard link. If the assist computer or camera dies, the MCU's radar warning and buzzer keep working. The MCU alone can show a minimal speed and warning on a small display.
Fog posts	Roadside boxes measure how far light can travel, plus temperature and humidity, and report whether fog is rising, steady or falling. We report current visibility and trend, not a 20-minute forecast.
Shared hazard list and messages	Trucks share small position-tagged messages carrying their age and a signature. Old or forged ones are dropped. Followers get early warning of obstacles they cannot see yet, shown as a Phantom Obstacle marker.
Blind-curve gates	Two gates at a single-lane blind stretch let one direction in at a time. If they are unsure, they say wait. [NEW] Lamp convention: flashing amber = hold, steady green = go.
Emergency beacon (optional)	[NEW] Carried by a blast supervisor or shift lead. One button sends a signed stop message that every truck shows as a critical alert, without going through a control room. Unsigned stop messages are rejected.

Messages travel over ESP-NOW (fast beacons, about 10 per second). LoRa carries only slow traffic: fog posts, the emergency beacon and long-range relay. The truck's radar loop depends on neither.

Why radar decides, even with four sensors

[NEW] At 3 to 5 m visibility every optical sensor loses range. Millimetre-wave radar is the least affected by fog and rain droplets, so it is the only sensor allowed to prove the road clear. Thermal, RGB and NIR add naming, extra detection and cross-checks, and can only make the truck more careful. LiDAR is not used: it is the most degraded by dense fog and the most expensive. Details on the next page.

Four sensors, one rule

Radar, RGB camera, thermal camera and NIR camera all feed the truck. They do different jobs, and none of them is trusted equally. The safety MCU only ever asks: what is the shortest distance at which any sensor says something is there?

Sensor	Job	Strength in fog	Honest weakness	Role in the safety decision
Radar (77 GHz in a mine; 60 GHz evaluation board or stand-in in the demo)	Range and speed of solid objects, day and night	Least affected by fog, rain and dust	Poor at naming objects; clutter from pit walls and reflections; coarse angle	Primary. The only sensor that can prove the road clear
Thermal (long-wave infrared)	Finds warm things: people, engines, tyres, running vehicles. Works in the dark	Sees through thin mist and dust better than visible light	Dense fog still shortens its range; cold rocks and debris show little contrast; low-cost modules have low resolution	Detects and names. Can shorten proven distance, never lengthen it
RGB camera	Names objects, reads road edges and the fade-test boards	Best on clear stretches and for the fog-post audit	In 3 to 5 m fog its range is about the visibility, so it is nearly blind	Assist and health checks. Shortens only
NIR camera (about 850 nm, with NIR lamp)	Extra contrast at dusk, night and in dust; reads NIR-reflective marker boards; compared with RGB for the fade test	Helps in dust and darkness	NIR light scatters in fog almost like visible light, so it does not see through dense fog	Assist and cross-check. Shortens only

Fusion rule (one-way).

1. Radar sets the proven-clear range.
2. If RGB, thermal or NIR reports an object nearer than the radar range, the proven distance shortens immediately, even if radar missed it.
3. "Camera sees nothing" is **never** proof. Fog blinds cameras, so no optical sensor can lengthen the proven distance or clear a radar hazard.
4. Two sensor types agreeing on the same bearing and range counts as a confirmed hazard for warnings. It still cannot clear anything.
5. Each sensor has its own health flag. A blinded sensor drops out and the corridor does not grow to compensate.

Matching is rule-based, not learned: detections are matched by bearing and range gates on the safety MCU side, and AI on the assist computer only names what the sensors found.

What each sensor catches

Situation	Radar	RGB	Thermal	NIR	What the truck does
Stopped dumper in mist	Sees	Late or no	Sees warm engine	Late or no	Shortens on first detection; two types agree, so confirmed
Person on the road	Weak	No	Sees	Weak	Thermal raises the hazard; radar range is checked
Fallen rock (cold)	Sees	No	Little contrast	No	Radar alone; the reason line says so
Mud or wood heap	May be weak	No	Low contrast	No	Residual gap: proven distance stays short, and the pass-through rule applies

Prototype hardware

Radar: 60 GHz evaluation board or labelled stand-in. RGB: Raspberry Pi camera. Thermal: a low-cost thermal array module (32x24 class) for the demo; a higher-resolution module for a real mine. NIR: a NoIR camera with an 850 nm lamp (850 nm rather than 940 nm, which most cameras see poorly). Planning add-on cost about Rs. 5,000 to 15,000 for the demo; verify prices and lead times. The thermal module is far too small for real mine range, so the demo proves the fusion rule, not mine-grade thermal range.

How one decision is made

1	Measure.	Radar tracks obstacles ahead and places them on the known route, so a truck on another branch is not mistaken for one on this road.
2	Prove.	Proven-clear distance is the shortest of three: the radar's healthy range, the nearest hazard or peer truck, and the first stretch nobody has proved (a blind curve without a gate OK, a map doubt, growing position error).
3	Need.	Needed distance adds driver reaction, braking and a buffer. Braking comes from tests on this class of truck at this load, slope and surface, always using the pessimistic value.
4	Compare and advise.	If proven is at least needed, the truck holds speed. If not, the screen shows a lower speed worked out from the same formula.
5	Show and log.	One speed, one arrow, one reason. Ten seconds before and twenty seconds after every warning are saved so a supervisor can replay exactly why. Near misses are also logged.

Distance needed = speed x reaction time + speed² / (2 x braking) + buffer.

Braking = tested brake value - slope effect - wet-road penalty.

Safe speed = braking x [-reaction + sqrt(reaction² + 2 x (proven - buffer) / braking)]. If proven is not more than the buffer, safe speed is zero: stop.

Worked example (illustrative numbers)

A loaded truck on a wet downhill drives at 30 km/h. Reaction time 1.5 s, buffer 10 m, braking 1.4 m/s² (2.5 minus 0.3 wet minus 0.78 slope).

Piece	Distance
Driver reaction (8.3 m/s x 1.5 s)	12.5 m
Braking (8.3 ² / 2.8)	25 m
Safety buffer	10 m
Truck needs about	47 m
Radar proves clear	30 m

47 m needed is more than 30 m proven, so the screen shows: "Slow to 20 km/h. Reason: only 30 m proven clear." With 40 m usable, an empty truck on a dry flat road (braking 3.0) may go about 42 km/h, while the loaded wet downhill truck may go about 31 km/h. Same radar range, different safe speed. A fixed "warn at 20 m" rule cannot do this.

[NEW] Worst case: what 3 to 5 m fog really means

Driver visibility of 3 to 5 m is not the same as proven distance. The driver cannot see 30 m, but healthy radar can prove it. The dangerous case is when proven distance itself collapses: a blind curve with no gate OK, a stale hazard, degraded radar, or heavy rain clutter. Same formulas, buffer 10 m, reaction 1.5 s:

Proven clear	Loaded, wet, downhill (braking 1.4)	Empty, dry, flat (braking 3.0)	Screen shows
100 m	50 km/h (site cap applies)	69 km/h (site cap applies)	Site speed cap
60 m	36 km/h	48 km/h	Advised speed
30 m	20 km/h	26 km/h	Advised speed
15 m	8 km/h	9 km/h	Creep, warn
10 m or less	0 (stop)	0 (stop)	STOP: not enough proven road

Reading the table honestly: when proven distance falls to the buffer, the truck stops and waits for proof (a passing truck, a gate OK, a supervisor). It does not creep blind. The productivity gain comes from the many stretches where radar proves 30 m or more; the safety gain comes from refusing to move when it does not. The site sets the buffer, and a smaller buffer at crawl speed is a site decision, not ours.

Blind curve: one direction at a time

Each gate asks each truck one question: how much room do you still have before the point where you can no longer stop? The truck with the least room goes first, because it cannot afford to yield. The other holds only if it can still stop.

Case	Room left A / B	Result
Loaded meets empty	6 m / 22 m	A goes, B holds
Empty truck already committed	18 m / -1 m	B goes, A holds
Gate message lost	any	Both wait. Uncertain is never clear. A truck already inside exits slowly on its own radar
Neither can stop in time	-4 m / -2 m	Both stop now. Logged as a design-level failure

Gate states: clear, reserved, occupied, uncertain, closed. A gate only clears after two independent signs the truck has left (gate sensor and the truck's own message).

Shared hazard list: cleared by looking, not by a timer

Step	What happens	Following truck sees
Spotted	Truck A's radar tracks something for several frames. It joins the list with a road position, unverified.	Possible hazard, approach with care
Confirmed	A second independent source (another truck's radar) sees it.	[NEW] Phantom Obstacle marker: hazard confirmed 120 m ahead, advised 12 km/h
Nobody looks	No truck has passed for a while. The record becomes stale, not clear.	Old report, approach slowly until someone looks
A truck passes	A healthy-radar truck drives through and sees nothing along the whole pass. Confirmed hazards need a second pass or a supervisor OK.	Being re-checked
Cleared	Removed because trucks actually looked.	Road proven clear, normal speed

Why not a timer? A timer would turn the hazard green after a few seconds of silence even if the rock is still there. Here silence only means "old report", and the next truck's radar heals it.

[NEW] Driver screen layout

The screen never shows a confidence percentage. Full driver-interface rules are on the next page. It shows one state, one speed, and the two distances that caused it, so the driver sees the reason at a glance.

SLOW DOWN	SAFE SPEED: 8 km/h	CLEAR 18 m NEED 25 m
Road strip: [YOUR TRUCK] ---- 18 m ---- [HAZARD marker] ---- (unproven)		

Example values only. Escalation is immediate; returning to normal needs stable health, a waiting period and driver acknowledgement.

Driver interface: eyes on the road, few alerts

Design principles. 1. The driver watches the windshield; the screen is a one-second glance. 2. An alert fires only when the driver needs to change something. 3. Unproven road is drawn as unknown, never as empty. 4. Nothing here can raise the safe speed.

1. Synthetic-vision road strip

Radar and thermal see what the driver cannot, so the screen draws it as a simple forward strip: the truck, road edges from the mapped route, and detected objects with distances. This extends the Phantom Obstacle marker so the strip is always drawn, not only when a hazard fires.

- **Objects:** up to the three nearest, each with distance and a small "seen by" tag (radar, thermal, peer).
- **Unproven stretch:** beyond the proven distance the strip is shaded "unknown". It never looks like empty road.
- **Position doubt:** when position error grows, the road edges fade and the uncertainty band widens. The optional lane cue (for example "left 1 m") is hidden when error exceeds the lane margin.
- **Glanceable:** no video, no 3D, one speed, one arrow, one reason. Above crawl speed the strip simplifies further.
- **Independence:** the safety MCU supplies the minimum layer (state, safe speed, nearest object distance). The assist computer adds the strip. If it dies, the screen falls back to the minimum layer.

2. Where the driver's attention goes

At 5 km/h a truck covers about 1.4 m each second, so a 2-second look at the screen is roughly 3 m driven without watching the road. In 3 to 5 m fog that is most of what the driver can see, and the windshield still shows road edges, people close by and cold debris that sensors may miss. So critical information must reach the driver without looking down.

Channel	Used for	Note
Sound	Short tone patterns and a brief spoken cue ("Slow down", "Hazard ahead, 80 metres")	Cabs are noisy, so sound is never the only channel
Touch	Seat or wheel vibration for critical alerts	Works without vision; needs at least a simple motor in the demo
Peripheral light	LED bar along the top of the dashboard: green, amber, red. A lit segment shows the side of the hazard	Seen without looking down; amber is a signal colour, not a claim about fog
Screen	Confirmation: state, safe speed, clear and needed distance, road strip	Mounted high, low-glare amber theme
Head-up display	Speed and warnings on the windshield	Future work; needs real hardware and calibration

Rule: every critical alert uses at least two channels.

3. Alert discipline against warning fatigue

A driver who mutes the system is worse off than one who never had it, and an experienced driver's instincts are tuned to what can be seen, which fog removes. So the aim is fewer alerts that each carry information the driver cannot get otherwise.

Level	Trigger	Output
Information	Fog rising, or proven distance changed, while the truck is at or below the safe speed	Screen colour or marker only. Silent.
Caution	Truck above the safe speed, or a new hazard confirmed ahead	One tone or one spoken cue, once
Critical	Cannot stop within the proven distance, hazard closing, radar degraded, signed emergency stop	Repeating tone, vibration and red LED bar

- **Speed-conditional:** the system speaks only when the truck exceeds the safe speed. Fog alone never triggers a sound.
- **One alert per hazard:** radar, thermal and a peer message reporting the same hazard produce one alert.
- **Repeat only on escalation:** a driver who complies gets a quiet screen; the alert re-fires only if the situation worsens.
- **Never suppress a first alert or an escalation.** Only repeats and duplicates are silenced.
- **Advisory wording:** information ("30 m proven clear"), not orders.
- **Dismissed-warning counter:** warnings that were ignored with no problem are logged by road segment, so false-alarm sources (a reflective wall, a dusty fog post, a bad map segment) are fixed before more warnings are added.
- **Return to normal:** stable health, a waiting period and driver acknowledgement, as before.

Honest limits. A hackathon cannot measure human factors. We show the rules working and claim no driver-trial result. Real tuning needs shadow-mode logging with drivers (Phase 3). Road-edge amber LED guides at blind-curve hotspots, water suppression of dust, and a head-up display are future work, not part of the prototype. Suppression rules depend on correct braking values, which is why the pessimistic value is used.

Fog posts get checked by the trucks that pass them

A dusty or drifting fog post could say "clear" while the road is white. So every passing truck estimates visibility itself and compares.

Topic	How it works
Camera fade test	Two high-contrast boards stand at known distances. The camera reads how faded each looks. Comparing the two cancels the unknown brightness and gives visibility of roughly 3.9 divided by the fading rate per metre (Koschmieder relation, 2% contrast threshold). Experimental, needs calibration.
Who wins a disagreement	Post says clear but trucks see dense fog: trucks win at once and the post is flagged for cleaning. Post says foggy but trucks see clear: the road stays cautious until several trucks agree. Mistakes fall on the careful side.
Fog or dust	Fog needs a small gap between air temperature and dew point. Bright scatter with dry air is tagged dust. Both cut what the driver sees, so the speed cap does not depend on the label.
Trustworthy messages	Each message carries age, sequence number and a signature. Age counters replace a shared clock, which a GPS-poor pit does not have. Old copies are rejected and logged. Messages that add caution are accepted when valid, so a forged one causes at most a nuisance slowdown. Messages that relax a restriction need stronger checks (see the security page).

When things break, the truck gets more careful

If this fails	The truck does
Network	Keeps radar and its own speed advice. Peer information goes stale and stops counting. Shows "cooperative link lost".
Camera or assist computer	[NEW] Safety MCU and radar continue unchanged. The road strip drops out and the screen falls back to minimum text: state, safe speed, nearest object distance. Alerts, vibration and LED bar keep working.
Thermal or NIR	[NEW] That sensor drops out. Radar and remaining sensors continue; the proven distance never grows to make up for it.
Fog post	Uses radar and the last known cap. A missing post never raises the speed.
Gate or its message	Gate says wait and trucks slow down.
Radar blocked	Shows "degraded, stop at a safe place". No other sensor pretends to replace it.
Fake or old message	Rejected and logged. Local warnings unaffected.

Warnings escalate at once. Returning to normal needs stable health, a waiting period and driver acknowledgement, so one good frame cannot cancel a real problem.

Security and trust

Principle. A jammed, forged or compromised link may slow a truck, but must never let it go faster than its own radar allows. So messages are trusted according to what they can do.

Message classes

Class	Examples	Trust rule
Adds caution	Hazard report, fog post says foggy, gate says hold, peer trucks nearby	Accepted if the signature and age are valid. A forged one causes at most a nuisance slowdown
Relaxes a restriction	Gate GO or exit clear, fog post says clear, hazard cleared, emergency-stop reset	Needs a per-device signature and a local check: a gate GO never raises speed above the truck's own radar limit; a hazard clears only when a truck actually looks; a fog-post clear needs trucks to agree; a stop reset needs a supervisor in person
Emergency stop	Beacon stop command	Signed with a key held by an authorised person, shown as critical on every screen, logged, revocable

Threats and defences

Threat	Defence	In the demo?
Forged or replayed message	Signature, age counter and sequence number kept in non-volatile memory so a reboot cannot reopen replays; rejects logged	Yes, 5:15
Forged relaxing message (fake gate GO, fake hazard cleared)	Class rule above; local radar check; gate exit needs two independent signs	Yes, 5:15
Radio jamming	Silence means uncertain: gates say wait, peer data goes stale, link-lost is shown. Radar loop does not depend on the radio	Yes, 4:30
GNSS jamming or spoofing	Position along the known route with IMU and wheel-distance cross-check; growing error shortens proven distance	Simulated
Radar blocked or interfered with	Health checks and a degraded state; no other sensor replaces radar. Interference from other radars is a field-test item	Yes, 5:15
Stolen or tampered device (truck unit, fog post, gate, beacon)	Production: per-device keys in secure elements and a revocation list. Prototype: shared demo keys, labelled	No
Malicious firmware update	Production: signed firmware, secure boot on the safety MCU, rollback protection	No
Beacon misuse	Per-holder keys, logging and revocation; a reset needs a person on site	Signed stop only

The ops dashboard is read-only by design and has no path into the truck safety loop, so compromising it cannot change a truck's speed advice.

Honest limits. The prototype uses a shared demo key, has had no penetration test, and claims no certification. Production key management is the hard part: enrolment, rotation, revocation and secure storage. A stolen valid device can still send validly signed lies, which is why relaxing messages need a second source or the truck's own radar. Production would follow ISA/IEC 62443 practice and a formal review; we do not claim compliance.

The six-minute demo

Time	What judges see	Proves
0:00	Two model trucks move. Screen green: "60 m proven clear, safe up to site cap".	Baseline
0:40	Mist starts. Fog post flags rising fog and advised speed drops before the truck arrives.	Early warning
1:30	A metal plate and a warm object are hidden in mist. RGB is blind; radar sees the plate, thermal sees the warm object. Each detection shortens the advice and the screen names the sensor ("Seen by: radar", "Seen by: thermal").	Radar-first proof, four-sensor rule
2:15	Switch to loaded and wet. Same distance, lower safe speed. [NEW] A truck above the new safe speed gets one tone; once it slows, the screen goes quiet.	Braking model, alert discipline
3:00	Truck A spots a hazard. Truck B behind sees [NEW] a Phantom Obstacle marker on its road strip, "hazard 80 m ahead, seen by peer". A truck passes, sees it gone, the list clears.	Shared hazard list
3:45	Two trucks meet at the blind curve. One GO, one HOLD.	Gates
4:30	Network off, then camera covered. [NEW] The safety MCU keeps the radar warning and buzzer running while the assist computer is dead.	Nothing fails silently
5:15	[NEW] Replay an old message and inject a forged gate GO: both rejected and logged, and the truck stays within its own radar limit. Block the radar: truck shows "degraded".	Honest failure, security
5:45	Replay screen shows why each decision was made, including a near-miss entry. Then the fleet simulation against "slow to a crawl".	Explainability, efficiency

The fleet simulation adds 10 to 20 virtual trucks on the laptop using the same messages. It is labelled a simulation and reports throughput against a "crawl in fog" policy using stated assumptions. **Freeze the scenario list first.** After that, add a feature only to fix a failure you measured.

What we build and what it costs

Item	Qty	Note
Model trucks: radar or labelled stand-in, IMU, wheel encoder, radios, screen, buzzer, safety MCU	2	Braking measured by stopping carts with weights
Sensor add-on: thermal array module, NoIR camera, 850 nm lamp	1 set per truck	Planning add-on about Rs. 5,000 to 15,000 per truck; verify prices
Driver interface: buzzer, vibration motor, short LED bar, small screen mounted high	1 set per truck	Planning add-on about Rs. 500 to 1,500 per truck beyond the screen and buzzer; verify prices
Fog posts: scatter sensor, temperature and humidity, LoRa	2	Battery or solar
Blind-curve gates: lamp and break-beam sensor	2	No central broker
Emergency beacon (optional)	1	Signed stop message, critical on every screen
Laptop dashboard, mist source, track, curve, obstacle, fog-check boards	1 set	Replay and analytics only
Planning total		About Rs. 50,000 to 95,000, or about Rs. 70,000 to 120,000 with one 60 GHz evaluation radar. Includes the thermal, NIR and driver-interface add-ons for two trucks. Estimates: verify prices and lead times.

Build priorities

Must build - Truck brain: safety MCU with measured braking - Radar and RGB, screen, buzzer - Alert logic (speed-conditional, one alert per hazard) - Message age, replay rejection and relaxing-message checks - Fog post with trend - Network, camera, radar failure demos	Then, in this order - Thermal into the corridor (shortens only) - Shared hazard list and road-strip display - Blind-curve gates - Fleet simulation - Camera fog check and NIR cross-check (drop first if behind) - Near-miss log entries in replay	Cut - 3D dashboard - Safety score - Worker tags - Escort mode - Head-up display and road-edge LED guides (future work) - RadarGlyph reflectors - Idle-truck observer nodes (simulation only) - Patent wording and any "100%" claim
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Flaws we have, said before you ask

"Matters" is our own judgement of how much each could hurt if ignored.

Flaw	What we do about it	Matters
Demo radar is a stand-in	It proves the logic, not mine-grade range. Mines need certified 77 GHz radar. We bench-test the stand-in on a stopped metal target from a moving cart and label it clearly.	High
Braking numbers are ours	They come from stopping our carts. A real mine must test its own trucks with the maker. We use the pessimistic value, and evidence can only lower the allowed speed automatically.	High
Radio in a pit is unproven	Bench shadow and reflect signals. We run range tests, let any truck relay once, and go careful when links degrade. Local radar never depends on it.	High
Gates could jam or false-clear	Uncertain blocks entry, exit needs two independent signs, and a manual clear procedure exists. Loss and delay cases are in the test matrix.	High
Trusting a leader's earlier pass	Letting followers go faster because a truck just passed could miss something that appeared afterwards. Off by default and simulation-only until field-tested.	High
Feature creep	The must-build list is fixed and nothing is added after the freeze except fixes for measured failures. v2 added no hardware.	High
[NEW] Fog this dense may leave little proven road	At 3 to 5 m visibility the truck depends entirely on radar range. If proven distance drops to the buffer the truck stops. That is correct behaviour but it costs cycle time; the fleet simulation shows how much.	High
[NEW] Four sensors is the biggest new scope	Fusion adds thermal, NIR, calibration and association code to a 36-hour plan. We keep the rule simple (extra sensors only shorten), build thermal first, and drop NIR first if behind. Thermal and NIR also lose range in dense fog, so they never clear a hazard.	High
[NEW] Alarm fatigue cannot be measured in a hackathon	We show the rules working (one tone, silence when compliant, escalation when not). Real tuning needs weeks of shadow-mode logging with drivers in Phase 3. We claim no human-factors result.	High
[NEW] Screen fixation and false confidence	A road strip can pull the driver's eyes off the windshield or make the road look clear. The strip is sparse, unproven road is drawn as unknown, and critical alerts use sound and vibration so the driver need not look down.	High
[NEW] Idle trucks as fog observers	A parked truck is often not on the fog-prone haul road and its sensors would need calibration. Kept as future work and simulation-only. We do not claim it in the demo.	Medium
Mist is not natural fog	Fog-machine droplets differ, so the sensor curve needs re-tuning on site. We claim repeatability, not mine equivalence.	Medium
Only two physical trucks	Fleet behavior comes from a labelled simulation on the same messages, not real fleet statistics.	Medium
No fog forecast	We measure current visibility and trend. Forecast probabilities need seasons of site data.	Medium
Prototype security is basic	Shared demo keys, no penetration test, no certification. Production needs per-device keys in secure elements, revocation, signed firmware and secure boot. A stolen valid device could still send signed lies, so anything that relaxes a restriction needs a second source or the truck's own radar. See the security page.	High
We advise, we do not brake	The demo relay is a labelled mock. Real intervention needs the truck maker's approval and functional-safety work (ISO 21815, ISO 19014). By design.	By design

Judge questions, answered plainly

Question	Answer
What is actually new here?	Every source of information has an age and a proof value. Only fresh proof lets a truck go faster, unknown never counts as clear, and hazards are removed by someone looking, not by a timer. Radar, V2V and fog sensors are not new alone; the combination and the rules are.
What if LoRa fails?	LoRa is not the safety link. Fast messages use ESP-NOW, and the truck's radar loop is independent of both. Lost links only make the truck more careful.
Why the loaded truck goes first?	It is harder to stop, and we work that out from distances instead of a fixed rule. If an empty truck has already committed, it goes first. A site rule can override.
Why not AI cameras?	Cameras degrade in fog. Radar decides, and AI only helps name what the radar found. The safety state is a formula you can replay.
Why four sensors, and which one decides?	[NEW] Each covers a gap: radar for range, thermal for warm objects, RGB for naming, NIR for dust and darkness. Only radar or fresh peer proof can lengthen the proven distance. The others can only shorten it, because fog blinds them. LiDAR is left out: it degrades most in dense fog.
Should the driver watch the screen or the windshield?	[NEW] The windshield. At 5 km/h a truck covers about 1.4 m every second, so a 2-second look down is about 3 m driven unseen. Critical alerts use sound and vibration, the LED bar is in the peripheral view, and the screen is a one-second glance.
What about warning fatigue?	[NEW] Alerts fire only when the truck is above the safe speed, once per hazard, and repeat only on escalation. A driver who complies gets a quiet screen. Dismissed warnings are counted so false-alarm sources can be fixed.
Does yellow light help in fog?	[NEW] Not by colour: fog scatters yellow and white about equally. We use amber only as a signal (gate lamps, low-glare screen) and never as a safety input. Radar decides.
What if the AI computer crashes?	[NEW] A separate safety MCU runs the radar loop, the formula, the watchdog and the buzzer. It does not need the AI computer.
What if GPS is wrong?	We track position along a known road with checkpoints, so GPS is optional. Growing position error shortens the proven distance and lowers the advised speed.
How does a driver see a hazard he cannot?	[NEW] The follower's screen draws a Phantom Obstacle marker at the hazard's road position with its distance, fed by a small position-tagged message, not video.
Who can stop all trucks?	[NEW] Only a signed emergency beacon carried by an authorised person. Unsigned stop messages are rejected.
Can it brake a real truck?	No. It is advisory. Real intervention needs the truck maker's interface, functional-safety analysis and mine approval.
Is it AI?	AI names objects and can help with fog trends later. The decision to slow down comes from the stopping formula.
How does it help productivity?	Clear stretches keep full speed and only risky segments are slowed. The fleet simulation reports the difference against slowing everything, using stated assumptions.
What if someone forges or jams messages?	[NEW] A forged hazard or hold message can only slow a truck. Anything that relaxes a restriction (gate GO, fog-post clear, hazard cleared) needs a per-device signature plus a local check, and never lets a truck exceed its own radar limit. Jamming makes gates say wait and peer data go stale; the radar loop is unaffected. The prototype uses shared demo keys and is not security-certified.

Where every idea came from

ProofRoad is a synthesis of eight documents: six original solution designs plus two later syntheses (AegisMesh and FogNet Sentinel). A seventh original design was mentioned but not received. Titles and names are as they appear in the submitted file names. **Check your event's attribution rules before submitting this page.**

Source document	What we kept	What we dropped
FogNet Nexus (Chatgpt_FogNet_Nexus.pdf, drafted with ChatGPT)	Green/yellow/red communication health. Movement-token idea. Stale-message rejection. Safety black box.	LoRa-only vehicle messaging. Alpha escort mode.
FogNet Alive (Claude_FogNet_Alive.pdf, drafted with Claude)	Passing trucks auditing fog posts. "A role, not an identity." Adaptive sampling. Emergency beacon. Stating limits before being asked.	Ultrasonic as main sensor. Signal-strength-only distance. Loaded truck always wins.
FOGNET 4.1 (AboliFogNet_4_1.pdf, by Aboli)	Explainable alerts. Safety states. Near-miss logging. "AI predicts, physics decides."	Worker tags and hazard beacons in the core.
FOG-SHIELD X (PrathameshFOG_SHIELD_X.pdf, by Prathamesh)	Shared hazard record with segment, severity, confirmations. Early warning for followers.	Invented percentage trust scores. Timer-based clearing. Mine Safety Index.
NEXORA FogNet SafeEnvelope (Snehal_NEXORA_FogNe.pdf, by Snehal)	Proven-clear corridor against stopping need. Measured braking bins. Fail-closed blind-curve block. Signed, age-checked messages. Test matrix. Honest claims.	Radar-reflector pattern decoding. Central broker. The larger budget and scope.
FogNet Swarm (GLM_Fogswam.docx, drafted with GLM)	Small position-tagged hazard packets shown on a following truck's screen. Mass-score right-of-way idea (replaced by room-to-stop).	Two-wavelength "100%" fog/dust claim. 3D fog cloud. Unauthenticated "Stop All".
[NEW] AegisMesh: Cooperative Kinetic Safety Architecture	Dedicated safety MCU tier. Phantom Obstacle presentation of shared hazards. Supervisor-carried emergency beacon. Idle-truck observer (as simulation-only future work).	RadarGlyph reflectors. Kinetic Priority score (units do not balance). IR/red fog-vs-dust claim. Worker tags. 77 GHz + RTK-GNSS cost base.
[NEW] FogNet Sentinel (FogNet_Sentinel_Final_Solution.pdf)	Driver screen layout with clear and needed distance. Safety Black Box with near-miss log. Local-safety-first hierarchy.	Nothing further; it is a subset of ideas already here.

New in ProofRoad: message freshness by age counters instead of a shared clock; hazards cleared by a truck passing; right-of-way from remaining room to stop; fog posts audited by the camera fade test; a virtual-fleet simulation for efficiency numbers; trusting a leader's earlier pass, held back as a simulation-only extra.

External references (as cited in the NEXORA dossier; not independently re-verified here)

1. Smart India Hackathon, official problem statements workbook (SIH26-26007).
2. ISO 21815-1:2022, Earth-moving machinery, collision warning and avoidance, general requirements.
3. ISO 19014-1:2018, Earth-moving machinery, functional safety methodology.
4. ISO 19014-2:2022, Safety-related hardware and architecture requirements.
5. EMESRT, Vehicle Interaction Control Improvement.
6. UK Met Office, How visibility and meteorological optical range are measured.
7. Sheeny et al., RADIATE: A Radar Dataset for Automotive Perception in Bad Weather, ICRA 2021.
8. Sezgin et al., Safe Autonomous Driving in Adverse Weather: Sensor Evaluation and Performance Monitoring, IV 2023.
9. Indian Railways, Basic Concept of Signal Engineering (block and one-train-only principles).
10. ISA, ISA/IEC 62443 industrial automation and control system security standards.

All numbers in this document are illustrative examples or planning estimates until measured. ProofRoad is advisory decision support, not a certified safety system. Claims should be read as targets or hypotheses until they pass testing.